Problem Set 2 - Linear Algebra [1]

Due date: 09/01

1. Let S be a sample space and let $\mathcal{B}$ be a σ-algebra on S. Use the properties of a σ-algebra to prove that:
   (a) $S \in \mathcal{B}$.
   (Hint: start with that $\mathcal{B}$ should be nonempty.)
   (b) $\mathcal{B}$ is closed under countable intersections.

2. Let $\mathbb{P}$ be a probability measure on a sample space S with σ-algebra $\mathcal{B}$, and let $A, B \in \mathcal{B}$. Prove the following properties:
   (a) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
   (b) $\mathbb{P}(A) \leq 1$
   (c) $\mathbb{P}(B \cap A^c) = \mathbb{P}(B) - \mathbb{P}(B \cap A)$
   (d) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
   (e) If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$

3. Let S be a sample space with σ-algebra $\mathcal{B}$, and let $A, B \in \mathcal{B}$. Prove that if A and B are independent, then the following pairs of events are also independent:
   (a) A and B^C
   (b) A^C and B
   (c) A^C and B^C

4. Let $\mathbb{P}$ be a probability measure on a sample space S with σ-algebra $\mathcal{B}$. Let $A, B, C \in \mathcal{B}$.
   (a) Show that $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$ does not imply that the events A, B, C are pairwise independent. (Hint: you only need to provide a counterexample).
   (b) What additional conditions are needed to guarantee that A, B, and C are mutually independent?

5. A test is used to detect the presence of a disease. The test has the following properties:
   - If a patient has the disease, the test always returns a positive result.
   - If a patient does not have the disease, the test returns a false positive with probability 0.005.

   Suppose the probability of having the disease is 0.001.
   If a patient receives a positive test result, what is the probability that they have the disease?

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6. A variable X is lognormally distributed if $Y = \ln(X)$ is normally distributed with mean μ and variance σ^2. That is, $f_Y(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2}$. Let the transformation be defined by $x = g(y) = e^y$ so that $y = g^{-1}(x) = \ln(x)$.
   (a) Derive $f_X(x)$.
   (b) Derive $\mathbb{E}[X^t]$ using $M_Y(t)$. What are $\mathbb{E}[X]$ and $V(X)$?

7. Let us consider the Law of Iterated Expectations in the continuous case. Suppose that $\mathbb{E}[Y] < \infty$. Prove the following results:
   (a) $\mathbb{E}[Y] = \mathbb{E}\Big[\mathbb{E}[Y|X]\Big]$
   (b) $\mathbb{E}[Y|X] = \mathbb{E}\Big[\mathbb{E}[Y|X,Z] \mid X\Big]$

8. Assume there are n volunteers elegible to receive a treatment. For each unit $i \in \{1, \cdots, n\}$, define the treatment indicator

$$D_i = \begin{cases} 1 & \text{if unit } i \text{ is treated} \\ 0 & \text{otherwise} \end{cases}$$

   Let $(D_1, \cdots, D_n)$ be the vector of the treatment indicators of all units. Due to capacity constraints, only $n_1 (< n)$ units can be treated: $\sum_{i=1}^n D_i = n_1$.
   (a) What is the total number of distinct treatment assignment vectors $(D_1, \cdots, D_n)$ we can construct?

   We say that treatment is randomly assigned if $(D_1, \cdots, D_n)$ are random variables, and if for any vector of n numbers $(d_1, \cdots, d_n) \in \{0,1\}^n$ such that $\sum_{i=1}^n d_i = n_1$,

$$P(D_1 = d_1, \cdots, D_n = d_n) = \frac{1}{\binom{n}{n_1}}$$

   That is, random assignment generates uniform treatment probabilities across units. Assuming the treatment is randomly assigned, answer the following:
   (b) For any unit $i \in \{1, \cdots, n\}$, what is $P(D_i = 1)$?
   (c) For any units $i \neq j$, what is $P\left(D_i = 1 \ \wedge \ D_j = 1\right)$? Is it true that unit i getting treated is independent from unit j getting treated?